

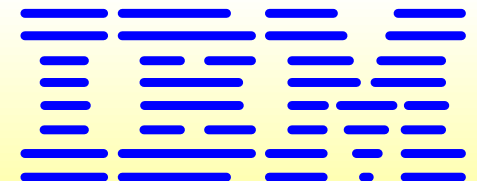


3019: DFSMS Basics: The ABC's of DFSMSrmm Part 2 of 2

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IBM Corporation
Warwick, UK

MikeW_Wood@UK.IBM.COM

A functional introduction to key RMM functions; how they work and what they deliver for your benefit. Learn how to define and manage the RMM resources and run the utilities



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Removable Media Manager

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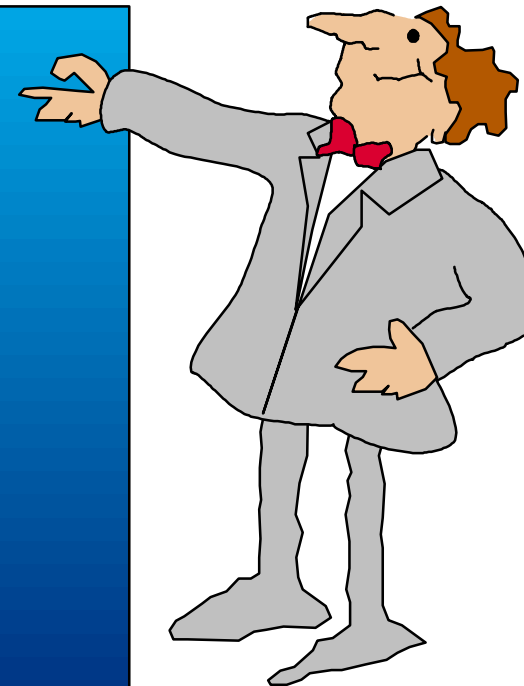
- DFSMSrmm
- IMS
- MVS
- OS/390
- RACF
- z/OS

Agenda for Part II



Removable Media Manager

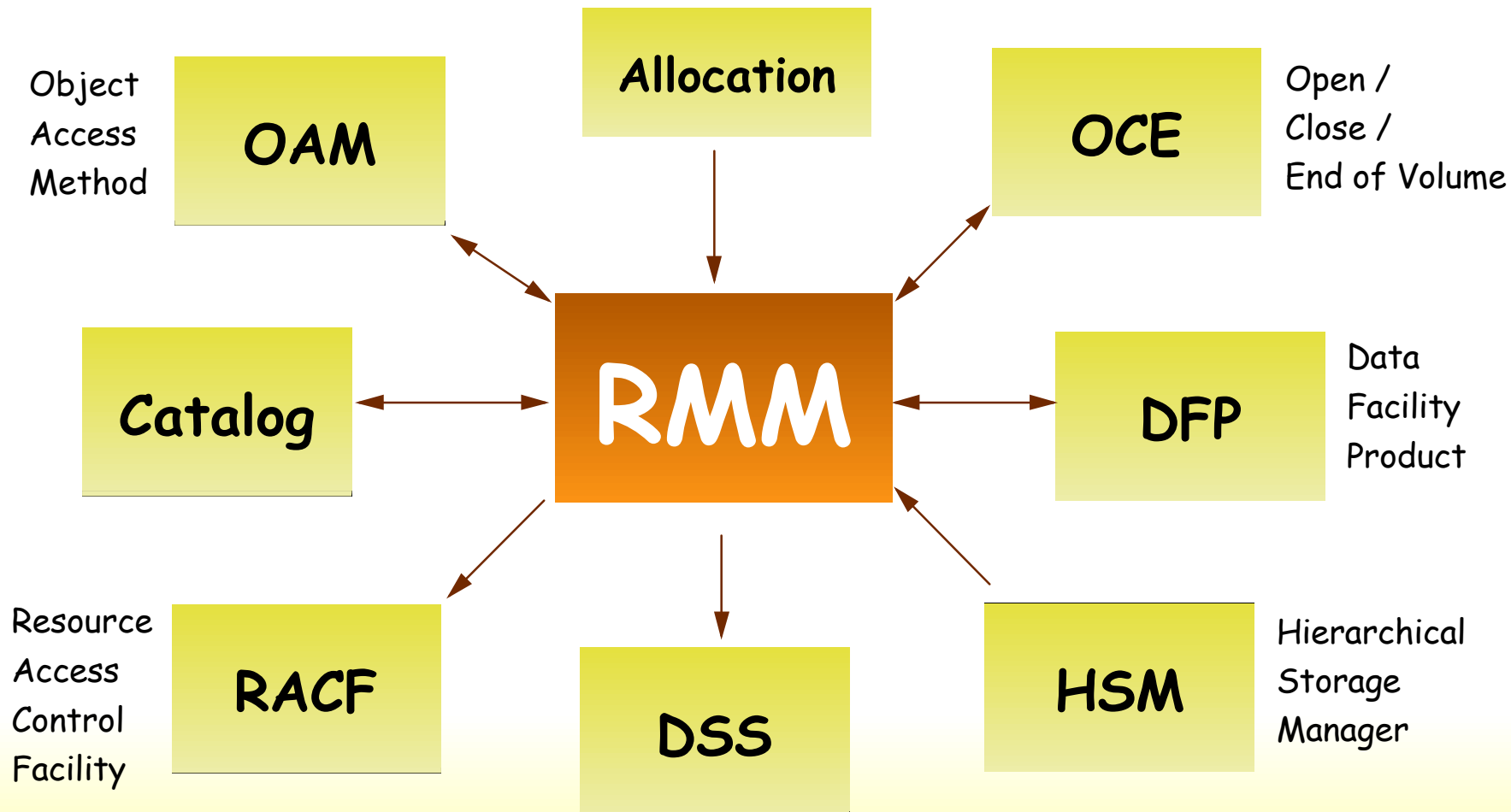
- Introduction to RMM
- Resources managed by RMM
- Software Interfaces
- User/Program Interfaces
- Inventory Management
- Tools



Software Interfaces ...



Removable Media Manager

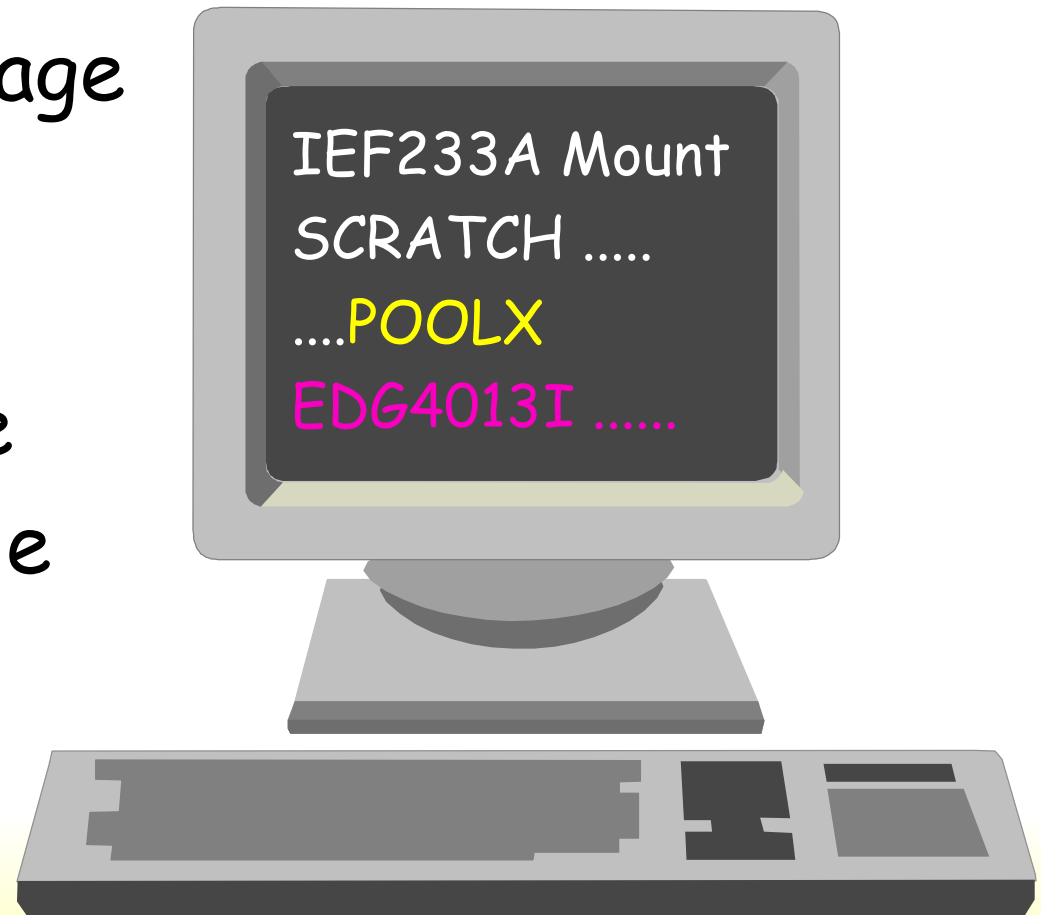


Allocation



Removable Media Manager

- Intercept Mount Message
- Assign Scratch Pool / Find Rack
- Modify Mount Message
- EDG4013I if unavailable



Open / Close / EoV



Removable Media Manager

- Validate Volume Use

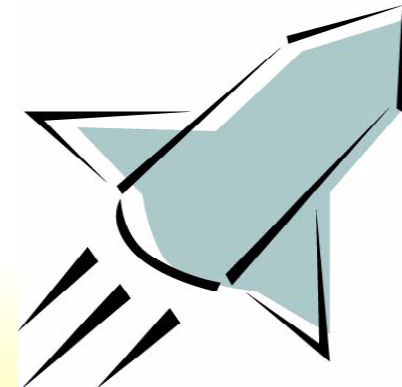


- Record Information in CDS for ...

- Volumes
- Data Sets



- Return Block-ID for High Speed Locate



Open / Close / EoV



Removable Media Manager

■ Reject Volume if ..

- Wrong volume mounted for a specific volume request
- Private volume mounted in response to a scratch request
- Attempt is made to read a scratch volume
- Attempt is made to overwrite a data set and the data set name does not match



Open / Close / EoV



Removable Media Manager

■ Reject Volume if ..

- An attempt is made to use a specific scratch volume
- The first file on the volume does not match what RMM has recorded
- An attempt is made to read or write a volume using nonstandard labels
 - Unauthorized function used
 - Bypass Label Processing
 - Write label, ...



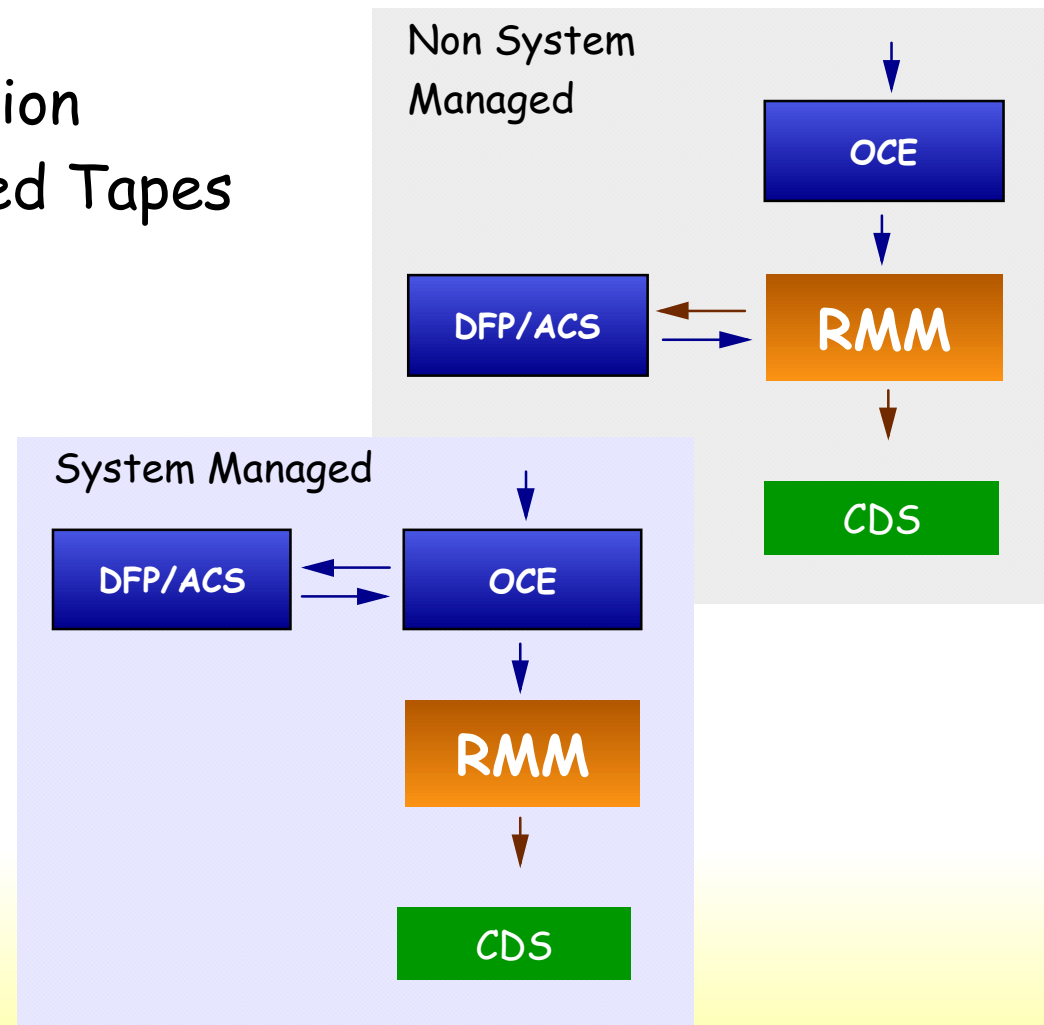
Data Facility Product



Removable Media Manager

- RMM calls
Automated **C**lass **S**election
for Non System-Managed Tapes
to get:

- Storage Group
for Volumes
- Management Class
for Data Sets

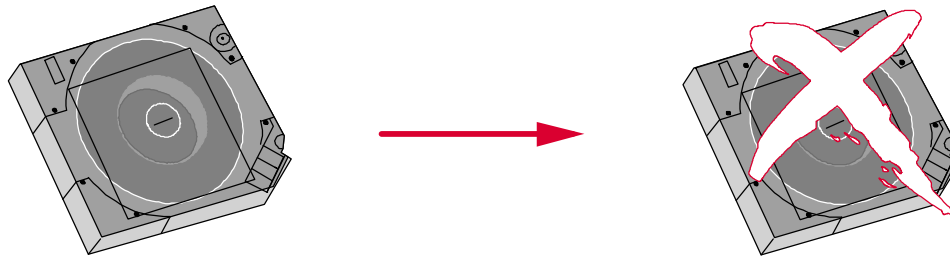


Hierarchical Storage Manager



Removable Media Manager

- HSM calls RMM to release a Volume



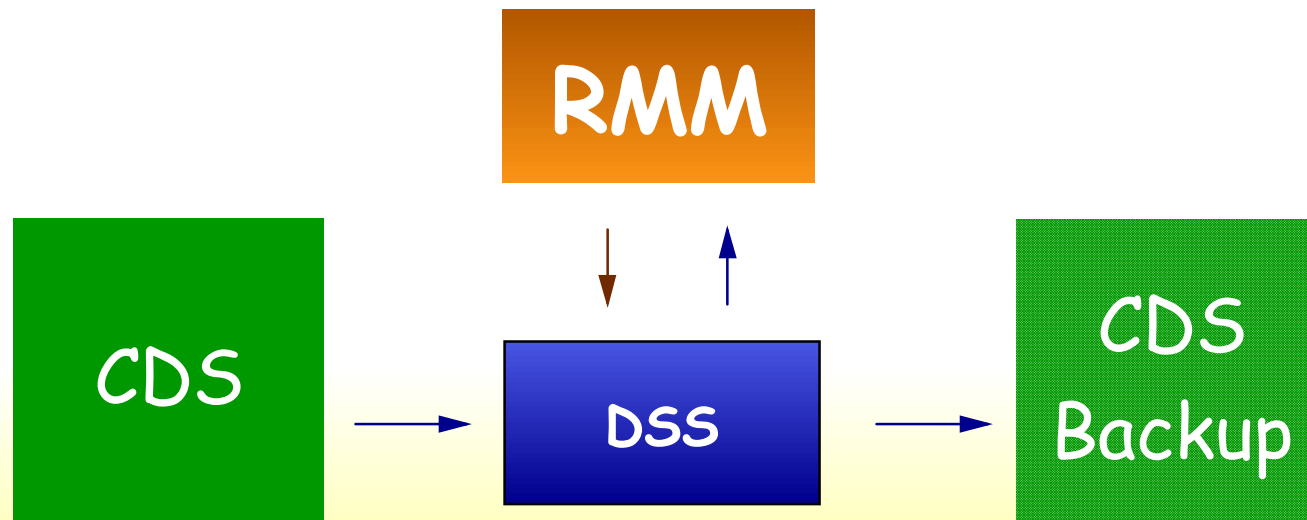
- Privileges for Volumes created by HSM
 - only check 17 Character Data Set Name
 - for input processing

Data Set Services



Removable Media Manager

- RMM invokes DSS for Non-Intrusive Backup of CDS
- RMM exploits DSS UIM



RACF



Removable Media Manager

- Protect RMM Resources
- Authorize RMM Functions
- Create and delete Tape Profiles
 - TAPEVOL
 - TAPEDSN

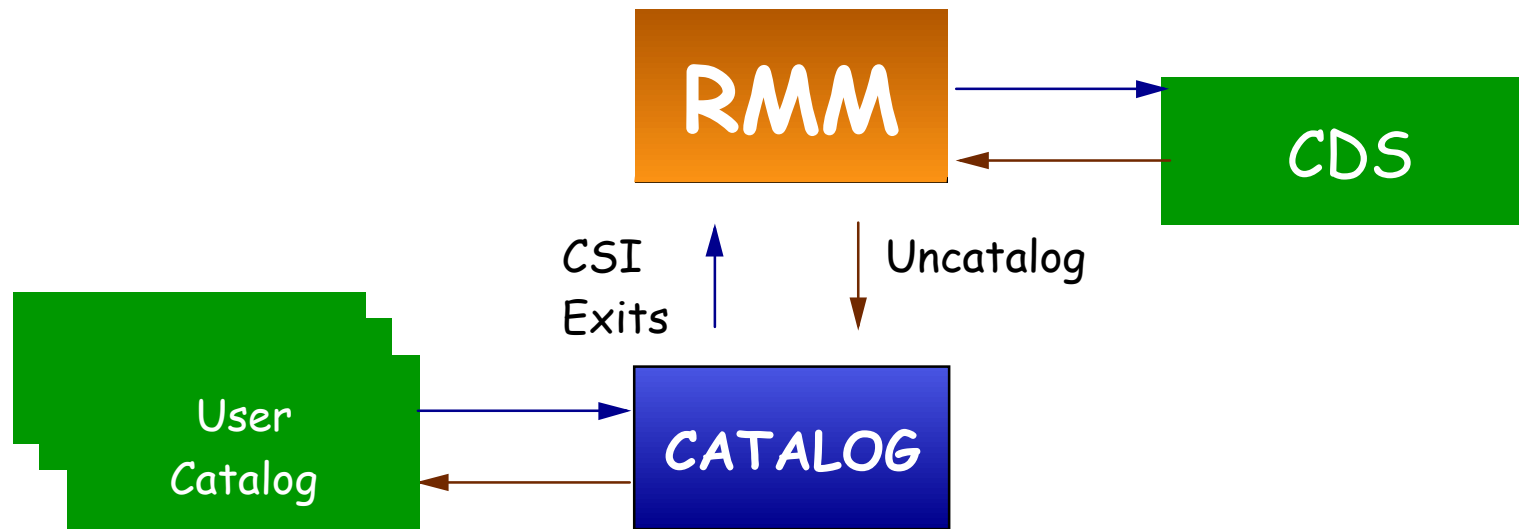


FACILITY:
STGADMIN.EDG.etc.

Catalog



Removable Media Manager



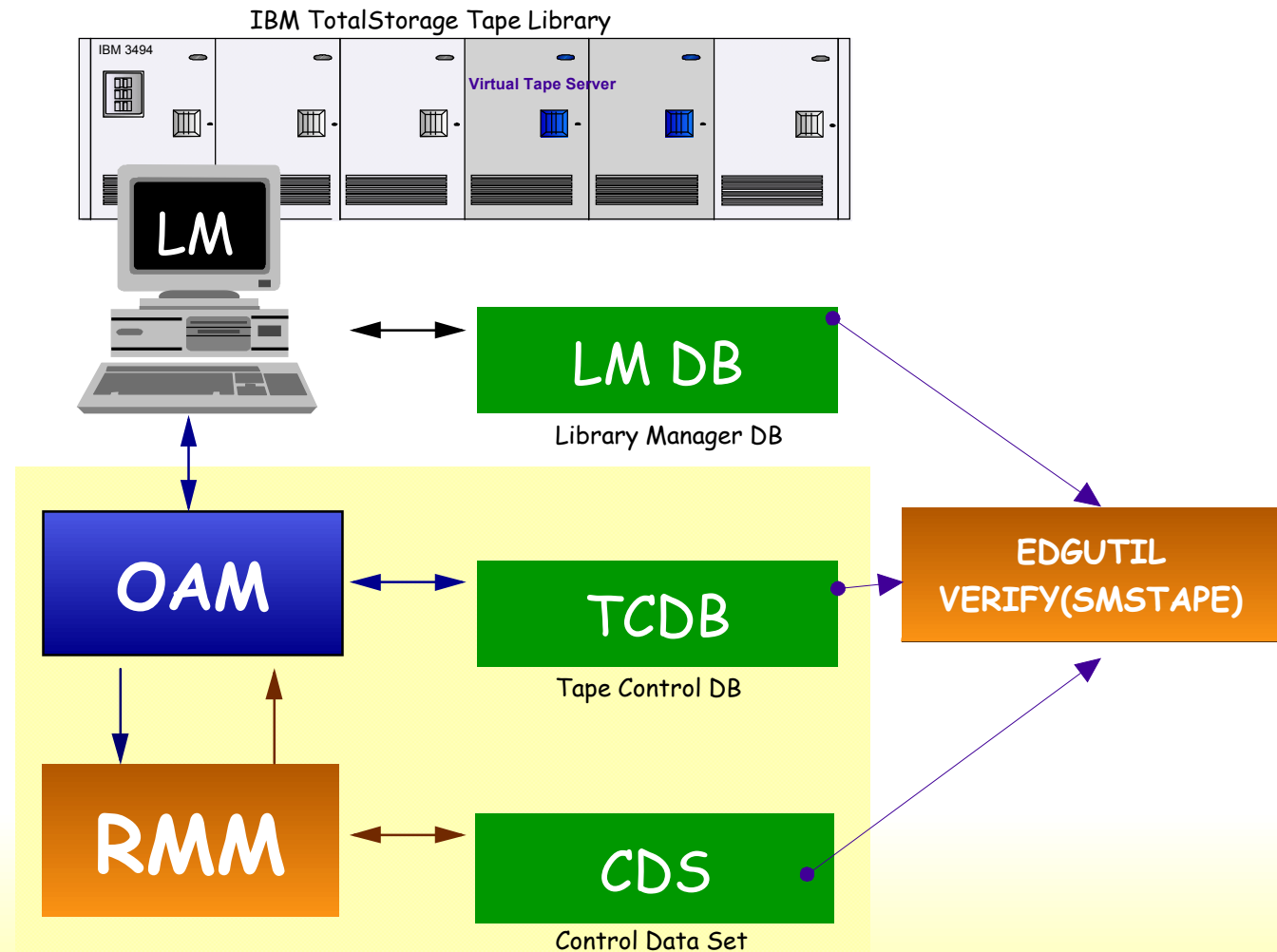
- Uses Catalog Search Interface to synchronize CDS with Catalog (tape data sets only)
- Exploits Catalog Exits to mirror Catalog updates in CDS
- Uncatalogs Tape Data Sets

Object Access Method



Removable Media Manager

- RMM's Interface to System Managed Tape Libraries



Object Access Method

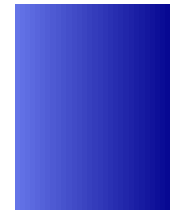


Removable Media Manager

< A 2-way Street >



- Volume entered into Library
- Volume ejected from Library
 - Volume not in Library ?
- Change Volume Use Attribute



- Query Library
- Query Volume
- Change Volume Status
- Eject Volume
- Delete Volume



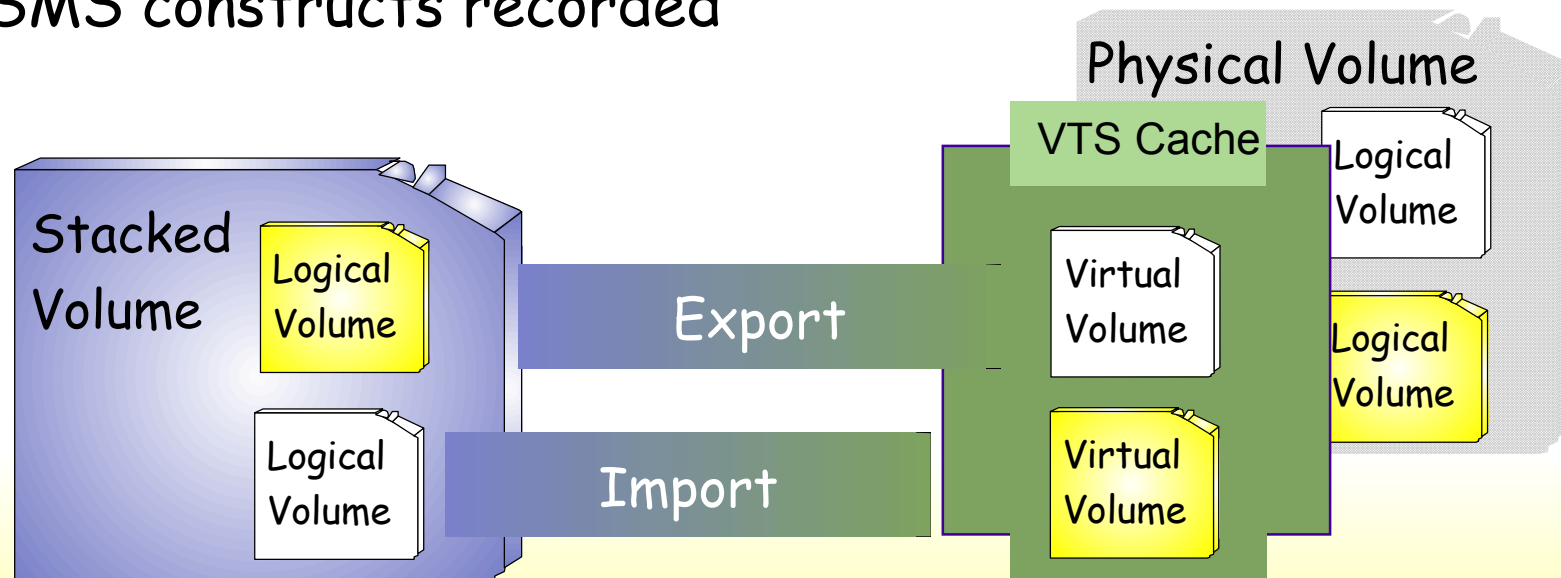
Object Access Method



Removable Media Manager

■ Virtual Tape Server

- RMM supports Export / Import
- Entry and Eject Exits used
- End Export message intercepted
- SMS constructs recorded



User interfaces ...



Removable Media Manager



↔ TSO Commands



RMM ADDVOLUME ...
RMM LISTDATASET ...
RMM SEARCHVOLUME ...
RMM CHANGEOWNER ...
RMM DELETEVRS ...

↔ ISPF Dialogs



DFSMSrmm User Menu

Option ==>

- 0 OPTIONS - Specify dialog options and defaults
- 1 VOLUME - Display list of volumes
- 2 DATA SET - Display list of data sets
- 3 PRODUCTS - Display list of Products
- 4 OWNER - Display or Change owner information
- 5 REQUE ST - Request a new volume
- 6 RELE ASE - Release an owned vo lume
- R REPORT - Work with reports

ISPF Dialog ...



Removable Media Manager

REMOVABLE MEDIA MANAGER (DFSMSrmm)

Option ==>

- | | | |
|---|---------------|---------------------------------------|
| 0 | OPTIONS | - Specify dialog options and defaults |
| 1 | USER | - General user facilities |
| 2 | LIBRARIAN | - Librarian functions |
| 3 | ADMINISTRATOR | - Administrator functions |
| 4 | SUPPORT | - System support facilities |
| 5 | COMMANDS | - Full DFSMSrmm structured dialog |
| 6 | LOCAL | - Installation defined dialog |
| X | EXIT | - Exit DFSMSrmm Dialog |

Enter selected option or END command.

For more info., enter HELP or PF1.

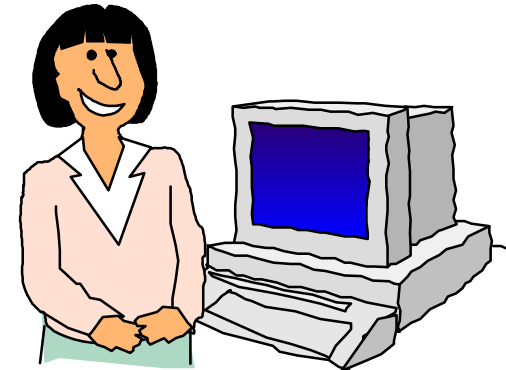
ISPF Dialogs ...



Removable Media Manager

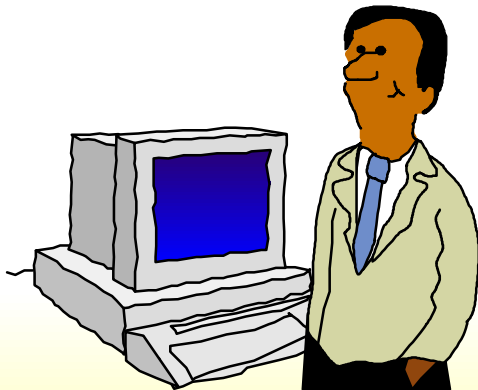
■ General User

- Request a scratch volume
- Change and release owned volumes / data sets
- Display most resources



■ Tape Librarian

- Define new volumes
- Add shelf locations
- Confirm volume movements and actions
- Report Generation



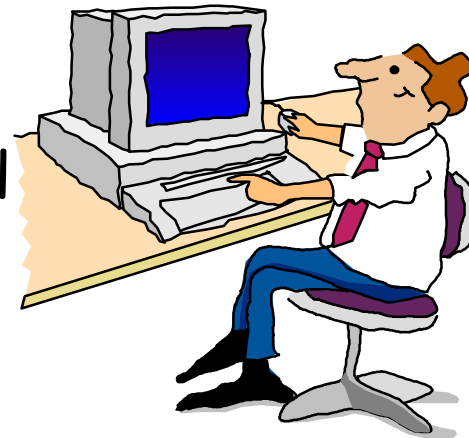
ISPF Dialogs ...



Removable Media Manager

■ Storage Administrator

- Add, Change and Delete Vital Record Specifications
- Report Generation



■ System Programmer

- Maintain parmlib options and CDS control information
- List software products

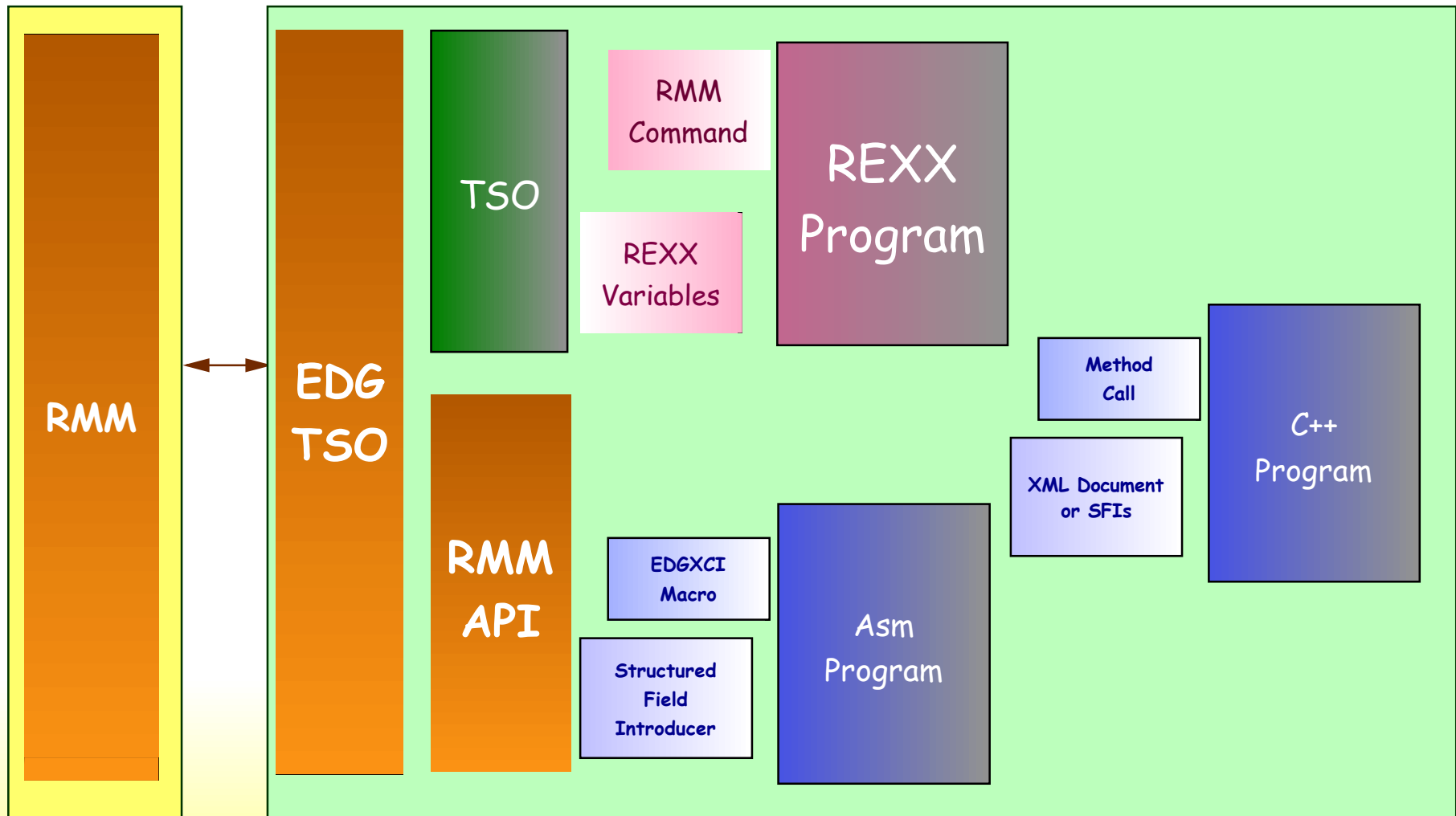
Programming interfaces ...



RMM Address
Space

Removable Media Manager

User Space



Inventory Management



Removable Media Manager

- Vital Record Selection
- Catalog Synchronization
- Expiration Processing
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup



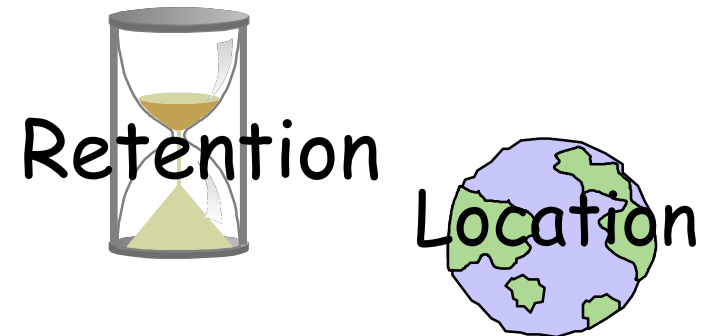
(Housekeeping
)

Inventory Management



Removable Media Manager

- **Vital Record Selection**
- Catalog Synchronization
- Expiration Processing
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup



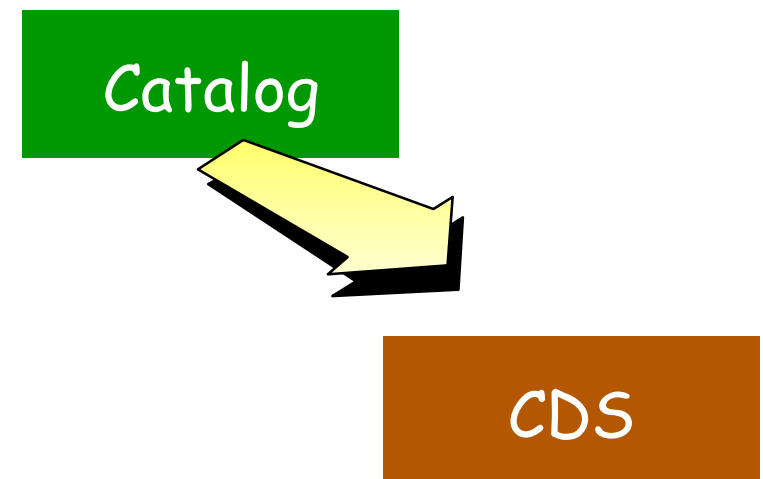
- Apply VRS Policies to
 - Data sets
 - Volumes
 - Volume Sets
 - Stacked Volumes
- Vital Record Report
- ACTIVITY File
- VERIFY run

Inventory Management



Removable Media Manager

- Vital Record Selection
- **Catalog Synchronization**
- Expiration Processing
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup



- Set Catalog Status for all Data Sets recorded in the CDS
- VERIFY
 - Report on Differences
- Run once to Implement

Inventory Management



Removable Media Manager

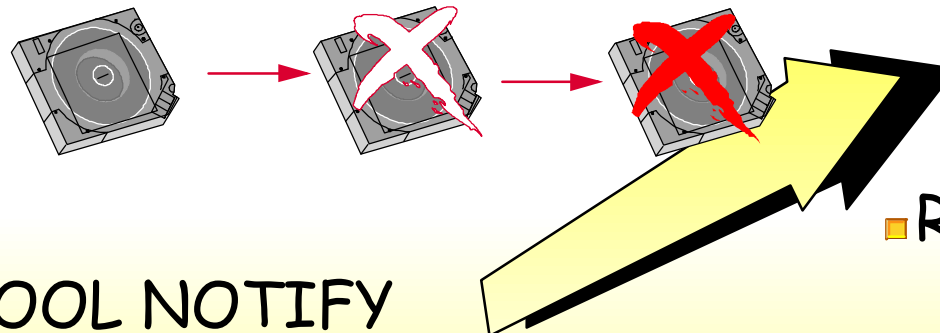
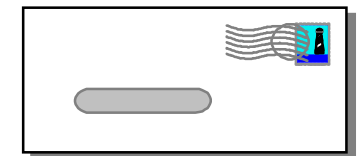
- Vital Record Selection
- Catalog Synchronization
- **Expiration Processing**
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup

- Release (Set Pending Release)

- Volumes
- Volume Sets
- VRS Release Options
 - Ignore EXPDT
 - Scratch Immediate

- Release Actions

- Notify Owner
- Replace/INIT/ERASE
- Return Volumes to Scratch
 - Manual Scratch option



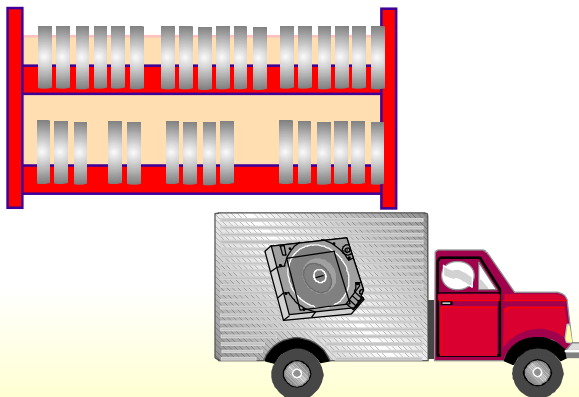
- VLPOOL NOTIFY

Inventory Management



Removable Media Manager

- Vital Record Selection
- Catalog Synchronization
- Expiration Processing
- **Storage Location Mgmt.**
- Report Extract
- Reports
- Backup
- Assign and release Bins
 - Optionally by Location
 - Reassign started moves
 - Optimized bin Assignment to Minimize slots required
 - Parmlib - REUSEBIN option
- Initiate/Confirm Moves

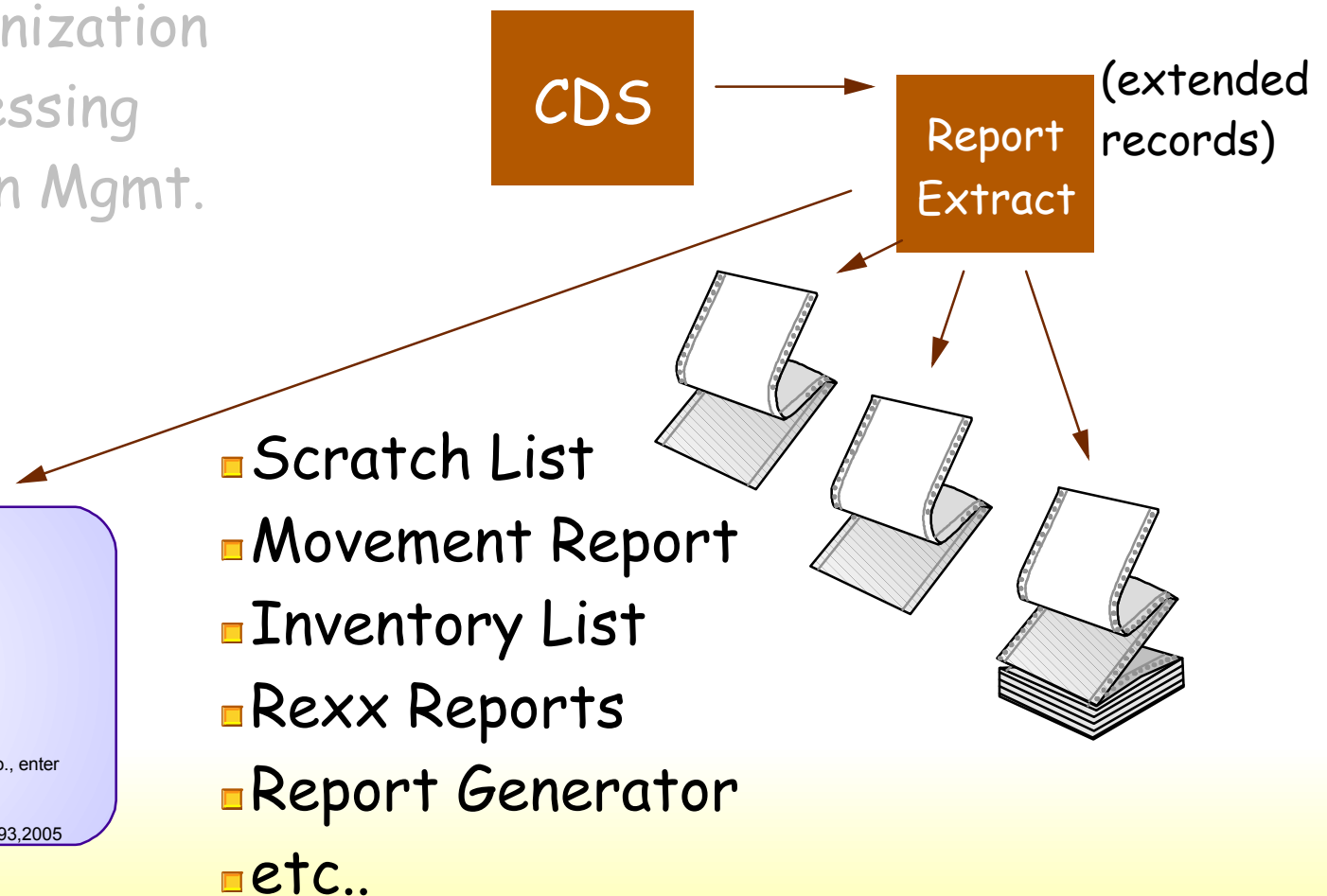


Inventory Management



Removable Media Manager

- Vital Record Selection
- Catalog Synchronization
- Expiration Processing
- Storage Location Mgmt.
- **Report Extract**
- **Reports**
- Backup



DFSMSrmm Report Generator
Option ==>>

- 0 OPTIONS - Specify dialog options
- 1 REPORT - Work with reports
- 2 REPORT TYPE - Work with report types
- 3 REPORTING TOOL - Work with reporting tools

Enter selected option or END command. For more info., enter
HELP or PF1.

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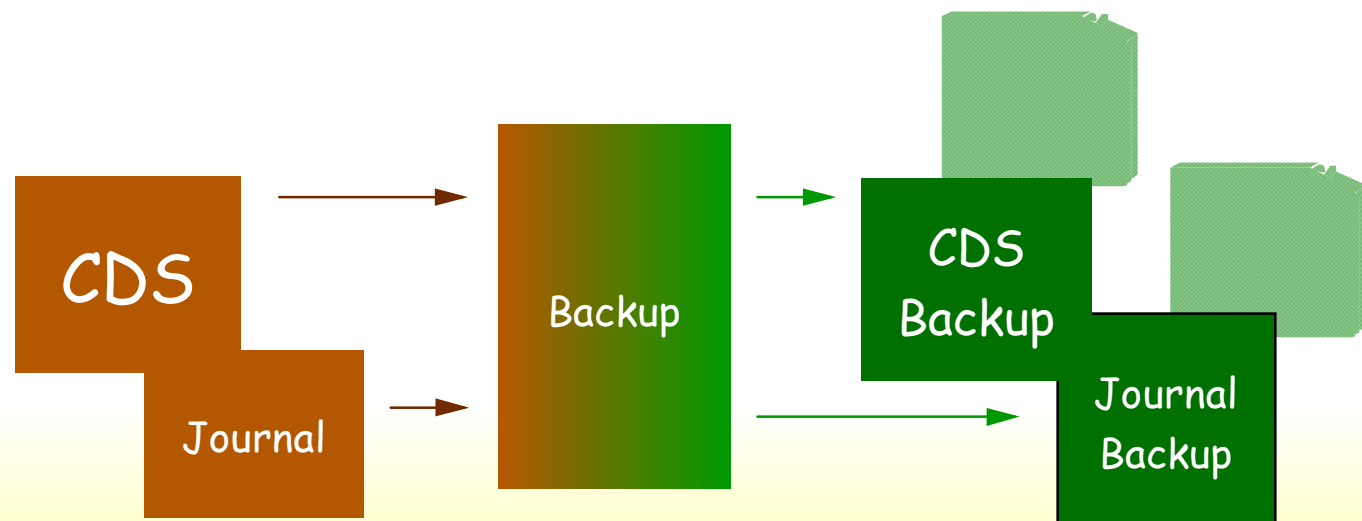
Inventory Management



Removable Media Manager

- Vital Record Selection
- Catalog Synchronization
- Expiration Processing
- Storage Location Mgmt.
- Report Extract
- Reports
- **Backup**

- CDS only
- CDS and Journal
- Journal Only
- Clears Journal

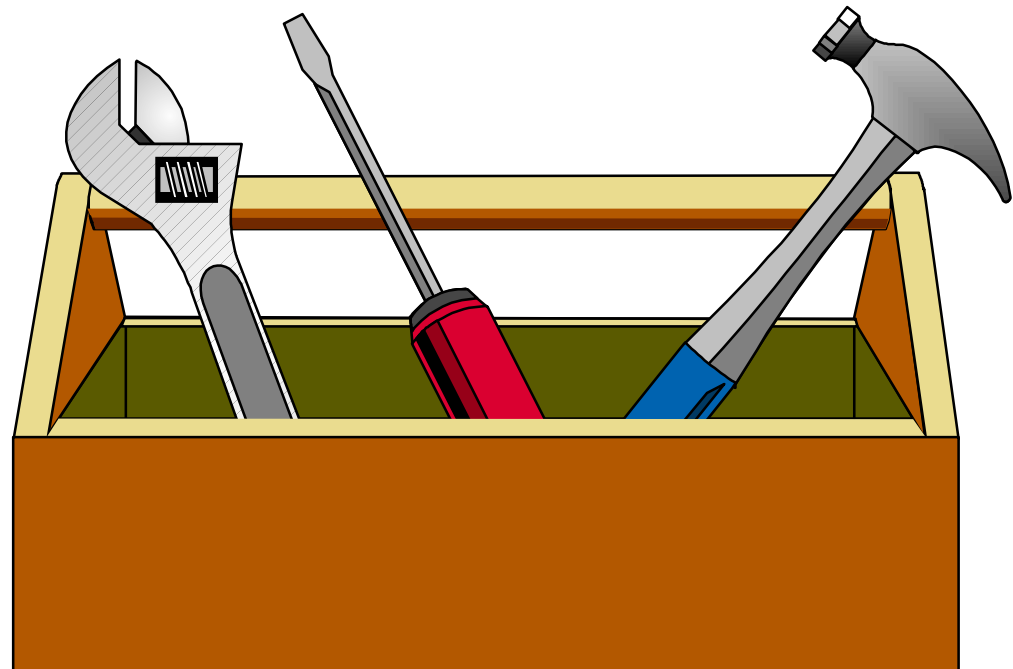


Tools



Removable Media Manager

- Reorg
- Recovery
- Verify / Mend
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- Conversion

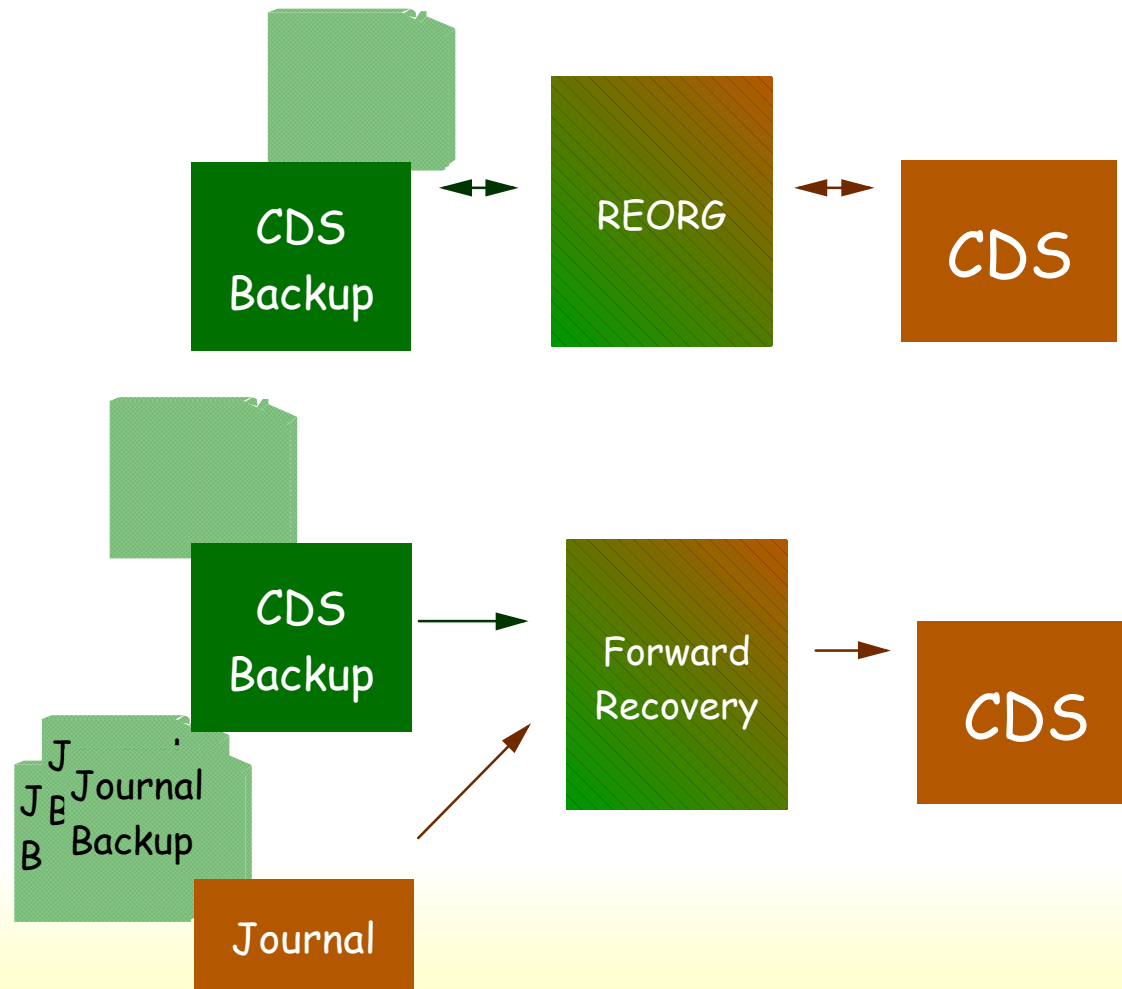


Tools



Removable Media Manager

- Reorg
- Recovery
- Verify / Mend
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- Conversion

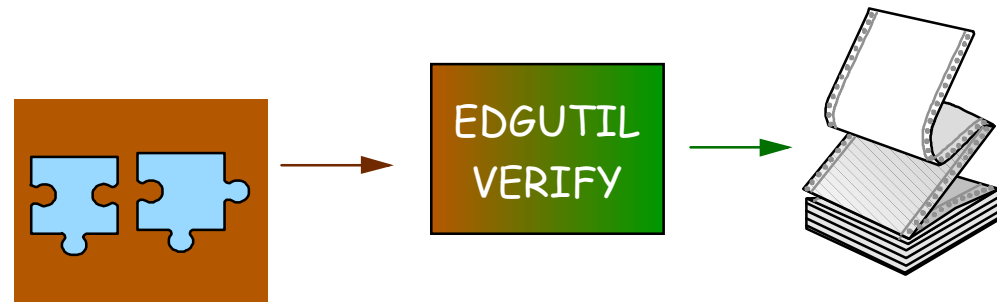


Tools



Removable Media Manager

- Reorg
- Recovery
- **Verify / Mend**
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- Conversion



- Report Inconsistencies in CDS



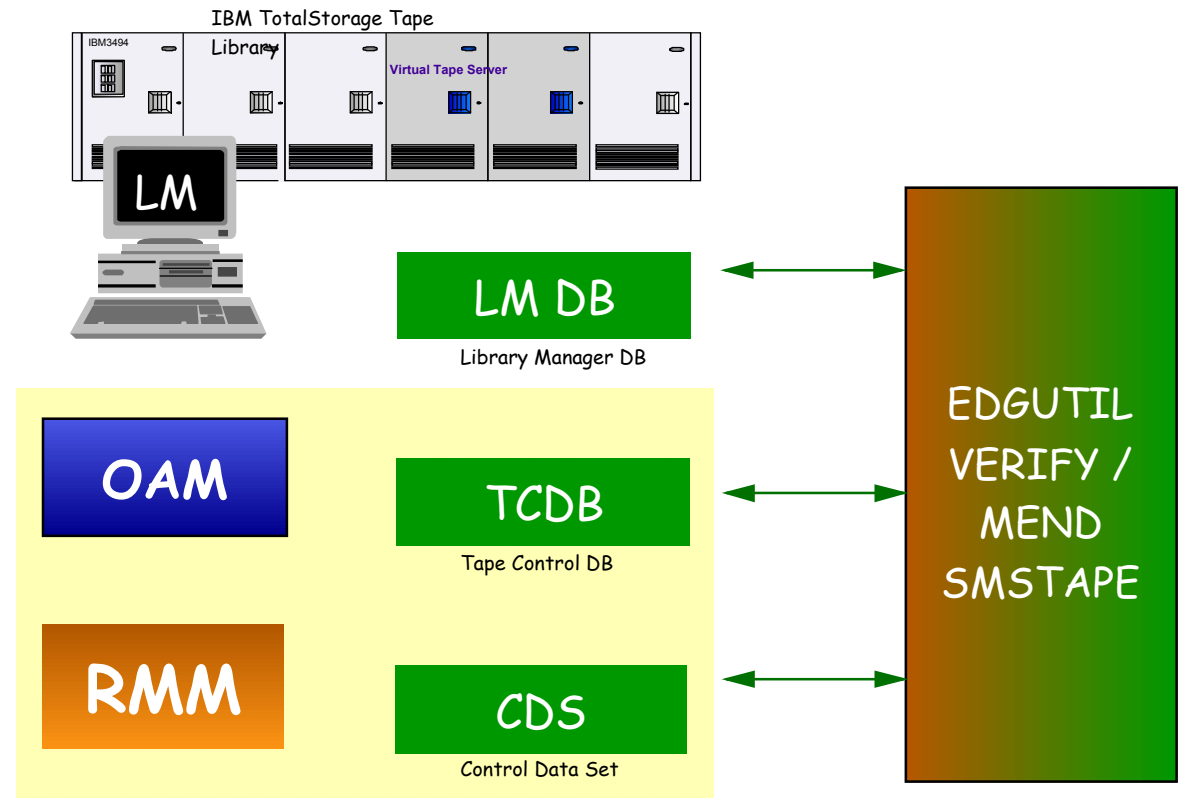
- Mend Inconsistencies in CDS

Tools



Removable Media Manager

- Reorg
- Recovery
- **Verify / Mend**
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- Conversion



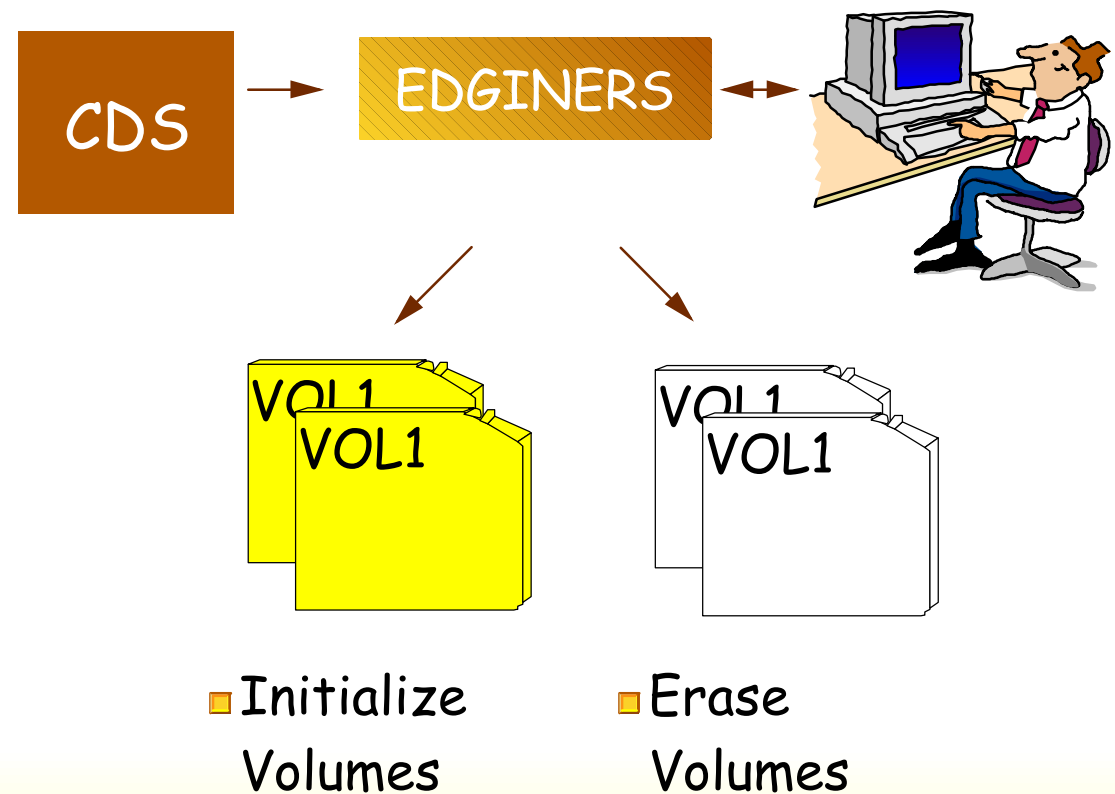
- 3-Way Audit

Tools



Removable Media Manager

- Reorg
- Recovery
- Verify / Mend
- **Init / Erase**
- SMF Reports
- PDA Trace
- IPCS
- Conversion

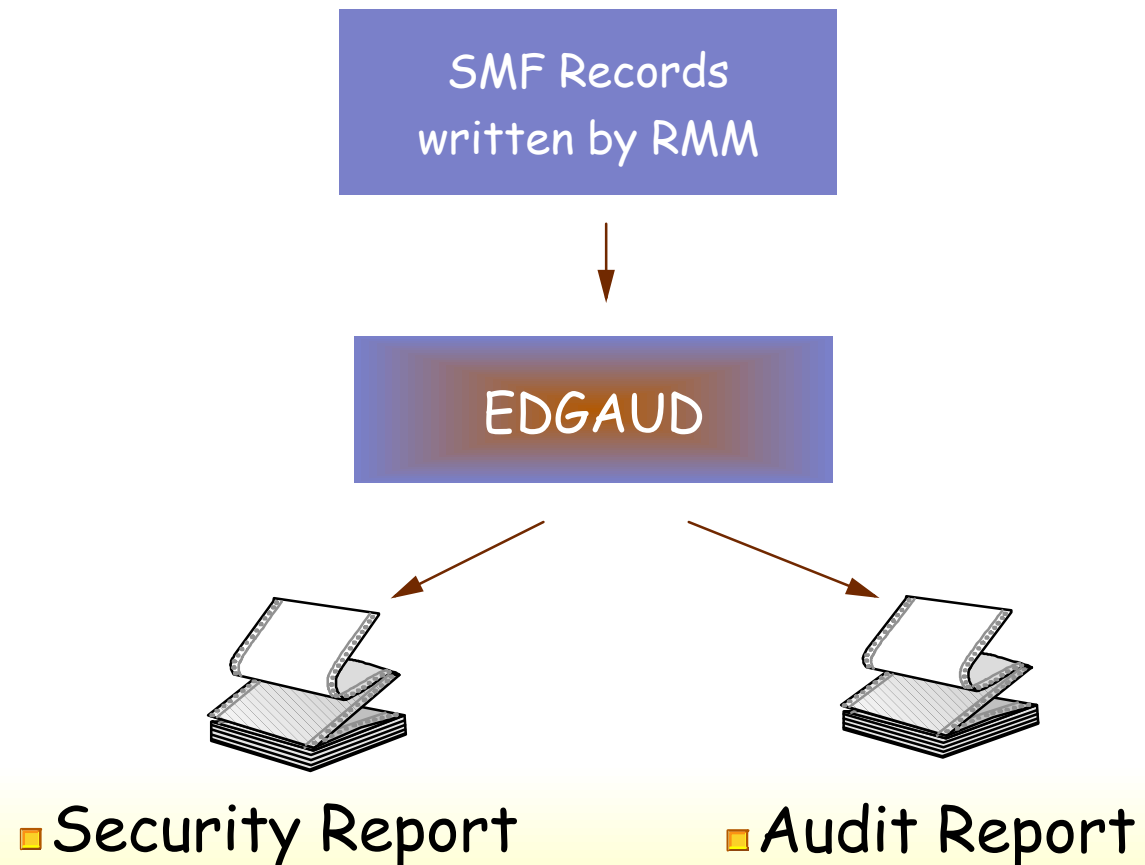


Tools



Removable Media Manager

- Reorg
- Recovery
- Verify / Mend
- Init / Erase
- **SMF Reports**
- PDA Trace
- IPCS
- Conversion



Tools

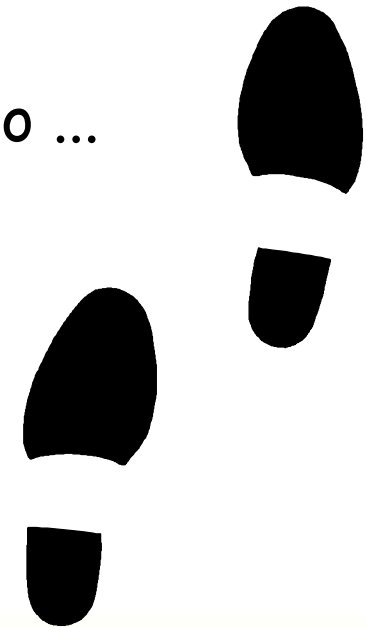


Removable Media Manager

- Reorg
- Recovery
- Verify / Mend
- Init / Erase
- SMF Reports
- **PDA Trace**
- IPCS
- Conversion

Problem Determination Aid

- Write Trace Records to ...
 - Storage Buffer
 - Data Sets
- Select and format Trace Records



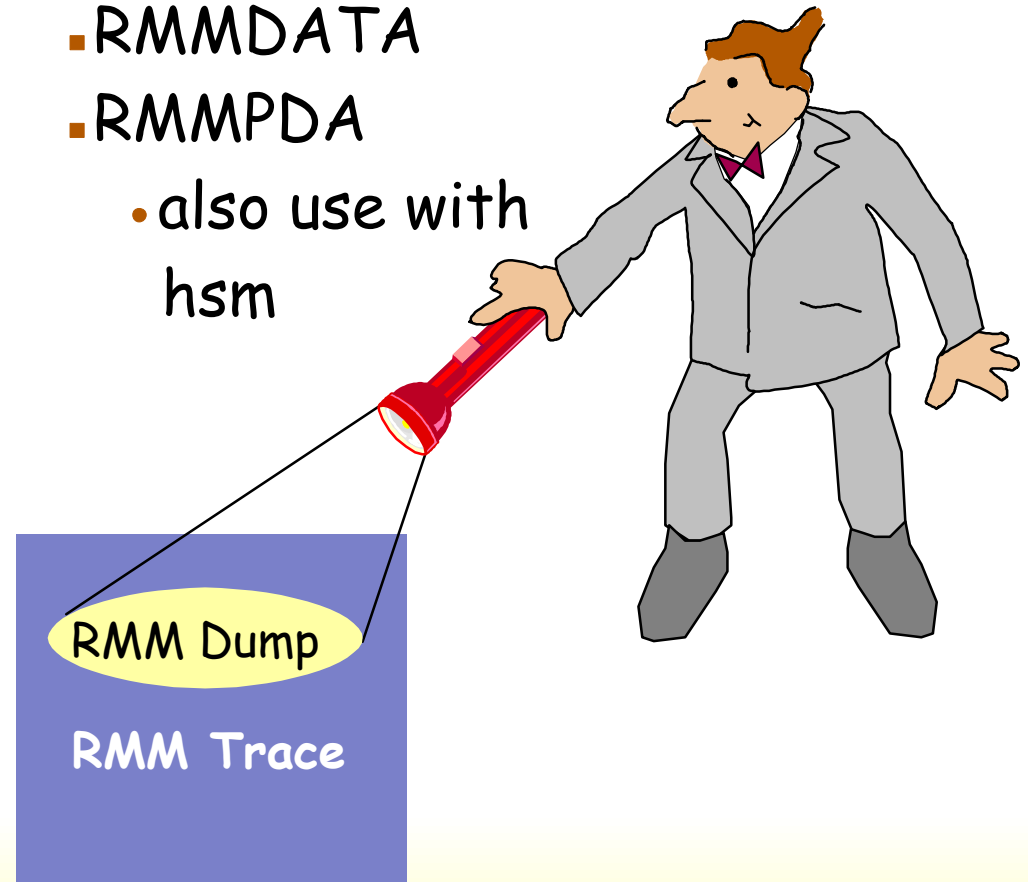
Tools



Removable Media Manager

- Reorg
- Recovery
- Verify / Mend
- Init / Erase
- SMF Reports
- PDA Trace
- **IPCS**
- Conversion

- IPCS verbexits
 - RMMDATA
 - RMMPDA
 - also use with hsm



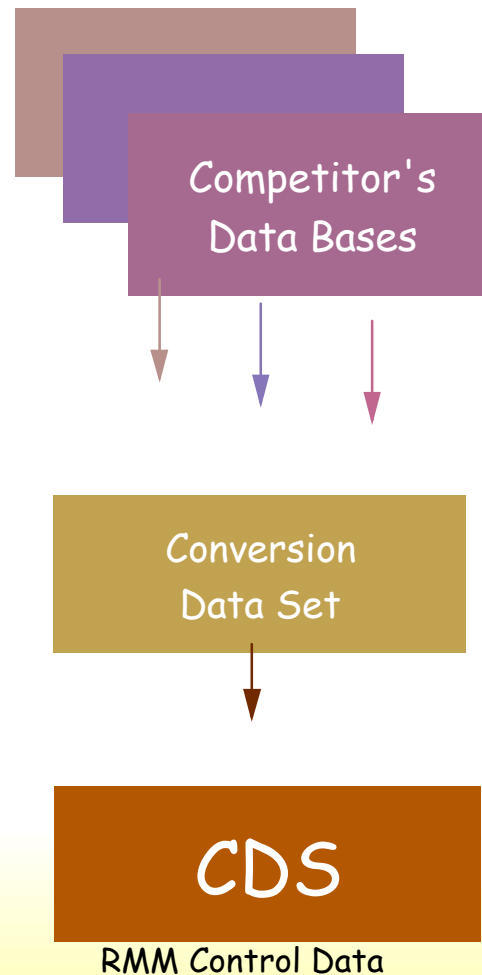
Interactive Problem Control System

Tools



Removable Media Manager

- Reorg
- Recovery
- Verify / Mend
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- **Conversion**



- IBM offers Migration Service
- Parallel run
- Mode of Operation
 - manual
 - record
 - warning
 - protect

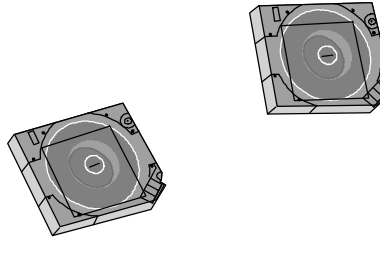
Summary



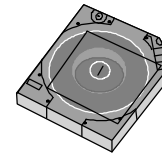
Removable Media Manager

Protect
Data on Tape

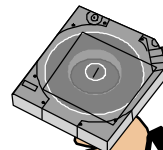
Enforce
Security



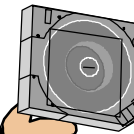
Control
Retention



Support
Automation



Manage
Vaulting



Thanks for your interest
in
DFSMSrmm ...

Removable Media Manager

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